

# Power Solar

*Multifunctional Solar Dryers and Rural Electrification*

## 01 PROBLEM

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### Context & Challenge

Power Solar is a company created in 2022 by Boni Sacca Tassigui, a renewable energy technician from a rural agricultural background. Products: multifunctional solar dryers for rural farmers, photovoltaic installations in rural areas, and maintenance. Innovation: IoT integration for mobile monitoring. Team: 3 permanent + 10 mobilisable technicians.

#### Core Problem Statement

Rural farmers lose 20–30% of their post-harvest yield due to inadequate drying. No competitor combines adapted rural technology, IoT monitoring, and maintenance service. However, Power Solar lacks a documented impact measurement system, the critical evidence required to attract NGO and development partner funding.

### Identified UX Problems

#### 01 Communication Centred on Needs, Not Achievements

Public communication reflects urgent funding need rather than accomplished results - undermining credibility with investors and NGOs who want evidence of performance.

#### 02 Single Revenue Source Business Model

After selling a dryer, there is no planned recurring revenue. Economic sustainability is not assured by one-time sales alone.

#### 03 Marketing Exposes Internal Processes

Facebook posts show internal manufacturing steps rather than the farmer's problem and social impact — risking competitor copying.

#### 04 No Formalised Impact Measurement

The team knows the project works from customer demand, but has no SMART indicator system to convince NGOs and development partners.

## Methodology

Research combined coaching sessions, competitive mapping across 3 actor types, and persona development from farmer and NGO program officer profiles encountered in the field.

## UX Competitive Analysis

| UX Criterion                     | NGOs / Grants | Traditional Dryers | Urban Suppliers | Power Solar          |
|----------------------------------|---------------|--------------------|-----------------|----------------------|
| Adapted to Rural Context         | Partial       | ✓✓                 | ✗               | ✓✓ Differentiator    |
| Dual Use (Drying + Electrif.)    | ✗             | ✗                  | ✗               | ✓✓ Key innovation    |
| Remote IoT Monitoring            | ✗             | ✗                  | ✗               | ✓✓ Field demand      |
| Local Maintenance Included       | ✗             | ✗                  | Partial         | ✓✓ Recurring revenue |
| User Training                    | Partial       | ✗                  | ✗               | ✓ In development     |
| Price Accessible Without Subsidy | ✓✓ (free)     | ✓✓                 | ✗               | ✗ → Key challenge    |
| Documented Measurable Impact     | Partial       | ✗                  | ✗               | ✗ → To build         |

✓✓ = *Very good*

✓ = *Present*

✗ = *Absent*

## User Personas

|                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
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| <p><b>Moussa K. - Rural Farmer</b><br/><b>Age:</b> 45 · Maize and groundnut farmer, North Benin · no electricity<br/><b>Goal:</b> Reduce post-harvest losses. Dry products quickly and cleanly.<br/><b>Frustration:</b> Dries on the ground. Loses 20–30% of harvest quality. Doesn't understand IoT.<br/><b>UX Need:</b> Simple interface (SMS or USSD), phone alerts on basic handset, dryer status without travel</p> | <p><b>Aminata S. - NGO Programme Officer</b><br/><b>Age:</b> 34 · Programme manager, agricultural development NGO<br/><b>Goal:</b> Identify a reliable local partner for post-harvest loss reduction in rural areas<br/><b>Frustration:</b> Suppliers don't document their impact. Can't justify funding without quantified evidence.<br/><b>UX Need:</b> Clear impact report, measurable SMART indicators, field references, structured quote</p> |
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## Key UX Insights

### **Insight #1 - Impact documentation is the unlock to NGO funding**

NGO officers cannot justify funding without documented SMART indicators. A downloadable impact PDF with concrete metrics (dryers installed, % post-harvest loss reduction, farmer testimonials) is the single highest-leverage asset Power Solar can create.

### **Insight #2 - Replacing 'IoT' with 'SMS alert on your phone' doubles comprehension**

Rural farmers understand 'SMS alert when your dryer needs attention' immediately. The word 'IoT' creates confusion and disengagement. Language must match the user's mental model, not the engineer's.

### **Insight #3 - Maintenance contracts create the missing recurring revenue**

After the dryer is sold, a maintenance subscription creates sustainable recurring income - transforming a one-time product sale into a long-term service relationship.

### **Insight #4 - Results-first communication replaces process communication**

Switching Facebook content from 'here's how we make the dryer' to 'this farmer saved 28% of his harvest this season' protects IP, builds social proof, and attracts both farmers and NGO partners.

## User Flows — 3 Critical Journeys

Friction points appeared consistently around raw material access, farmer purchasing power, and the absence of impact documentation for NGO validation.

**Journey 1 — Farmer/Cooperative** Discovery (field/word of mouth) → On-site demo → Purchase agreement → Installation + training → IoT mobile monitoring → Maintenance contract

**Journey 2 — NGO/Funder** Discovery (LinkedIn/network) → Impact file consultation → Negotiation meeting → Funding agreement → Field deployment → Evaluation report

**Journey 3 — Power Solar Operator** Order received → Prototype manufacturing → Field installation → IoT activation → Monitoring & maintenance → Invoicing & impact report

## Design Decisions

Based on research and coaching insights, the following UX solutions were designed covering a farmer-problem-centred homepage, a dual-audience solution page, and a monitoring & evaluation dashboard.

## Prototype :

<https://zadya22.github.io/PowerSolar/>

## User Testing Protocol

**Participants:** 2 rural farmers + 1 NGO programme officer + 1 internal operator

**Duration:** 30 min per session

| #  | Task                                                                                                           | Focus Area                      |
|----|----------------------------------------------------------------------------------------------------------------|---------------------------------|
| T1 | You are a farmer. Understand the difference between the standard dryer and the IoT dryer, and request a quote. | Product differentiation & quote |
| T2 | You are an NGO programme officer. Find the deployment pack and download the impact file.                       | NGO discovery & impact proof    |
| T3 | You are Boni (operator). Check the Ouidah cooperative dryer status and schedule a maintenance visit.           | Operator IoT monitoring         |
| T4 | Find Power Solar's impact numbers (dryers installed, loss reduction, cooperatives).                            | Impact metrics discoverability  |
| T5 | From the results chain, explain the difference between an output and an outcome for Power Solar.               | M&E framework comprehension     |

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## Test Report - Key Findings

|                                                                   |                                                                                                                                                                                                                                               |
|-------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>T1 - Dryer Quote (Farmer)</b><br><b>Success: 3/5</b>           | Farmers couldn't distinguish the two dryers. The IoT mention created confusion. Recommendation: Replace 'IoT' with 'SMS Alert on your phone'. Use field language, not technical language.                                                     |
| <b>T2 - Impact File (NGO)</b><br><b>Success: 2/5</b>              | The programme officer found no downloadable impact report. The page mentions figures but provides no structured document. Recommendation: Create an 'Impact' page with a downloadable PDF containing SMART indicators and field testimonials. |
| <b>T4 - Impact Metrics</b><br><b>Success: 4/5</b>                 | Metrics are present on the homepage but scattered. The user must scroll to find them all. Recommendation: Create a dedicated 'Our results in numbers' block visible without scrolling. Max 4 indicators.                                      |
| <b>T5 - Output vs. Outcome (Dashboard)</b><br><b>Success: 5/5</b> | No issues. The results chain was understood thanks to concrete examples. Recommendation: Keep this format. Add a printable infographic for NGO meetings.                                                                                      |

## Next Steps

- Create a downloadable impact PDF with SMART indicators and farmer testimonials
- Switch Facebook content to results-first format: farmer story + % losses saved
- Draft maintenance subscription contract for post-sale recurring revenue
- Build the monitoring dashboard to generate quarterly impact reports automatically